

# Vidya Sagar Sharma

Institute Postdoctoral Fellow, Department of Computer Science and Engineering, IIT Madras

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## Research Interests

Parameterized algorithms and complexity; graph algorithms; combinatorial optimization; causal graphical models; Bayesian networks; counting and sampling algorithms.

## Positions

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| <b>Apr 2025–Present</b>  | <b>Institute Postdoctoral Fellow</b> , IIT Madras. Joined 9 April 2025. Mentor: Dr. Akanksha Agrawal.                                                                 |
| <b>Aug 2024–Mar 2025</b> | <b>Senior Project Officer</b> , Centre for IC&SR, IIT Madras. Project: <i>Parameterized Algorithmics–Tradition &amp; Beyond</i> . Supervisor: Prof. Akanksha Agrawal. |
| <b>Aug 2014–Jul 2015</b> | <b>Associate System Engineer</b> , IBM India Pvt. Ltd.                                                                                                                |

## Education

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| <b>2017–2025</b> | <b>PhD in Computer Science</b> , TIFR, Mumbai. Advisor: Prof. Piyush Srivastava. Thesis: <i>Counting Problems in Graphical Models</i> . Submitted January 2025; defended 31 July 2025; degree awarded November 2025. |
| <b>2015–2017</b> | <b>M.Tech. in Theoretical Computer Science</b> , IIT Guwahati. Advisor: Prof. Deepanjan Kesh.                                                                                                                        |
| <b>2010–2014</b> | <b>B.Tech. in Computer Science and Engineering</b> , NIT Patna.                                                                                                                                                      |

## Publications

1. J. Harviainen, P. Parviainen, and **V. S. Sharma**. “Learning Bayesian and Markov Networks with an Unreliable Oracle.” *UAI 2026*, PMLR 337, 2026.
2. **V. S. Sharma**. “A Fixed-Parameter Tractable Algorithm for Counting Markov Equivalence Classes with the Same Skeleton.” *AAAI*, 2024. arXiv:2310.04218.
3. **V. S. Sharma**. “Counting Background Knowledge Consistent Markov Equivalent Directed Acyclic Graphs.” *UAI*, 2023. arXiv:2206.06744.
4. D. Kesh and **V. S. Sharma**. “On the Bitprobe Complexity of Two Probe Adaptive Schemes.” *Discrete Applied Mathematics* 319 (2022), 132–140; extended abstract in *CALDAM 2019*.

## Preprints

- **V. S. Sharma**. “Faster Fixed Parameter Tractable Algorithms for Counting Markov Equivalence Classes with Special Skeletons.” arXiv:2312.17626.
- **V. S. Sharma** and P. Srivastava. “The PSPACE-Hardness of Understanding Neural Circuits.” arXiv:2006.08266.

## Selected Talks and Presentations

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- “Counting Problems in Graphical Models,” invited talk, Modern Trends in Parameterized Algorithms, IIT Madras, December 2025.
- “A Fixed-Parameter Tractable Algorithm for Counting Markov Equivalence Classes with the Same Skeleton,” AAAI 2024, Vancouver, Canada.
- “Counting Background Knowledge Consistent Markov Equivalent Directed Acyclic Graphs,” UAI 2023; STCS Annual Talks, TIFR; TIFR Open Seminar.
- “On the Bitprobe Complexity of Two Probe Adaptive Schemes Storing Two Elements,” CALDAM 2019, IIT Kharagpur.
- Posters on Minimum Consistent Subset and shortest-cycle problems, IIT Madras Poster Day, February 2026.

## Research Visit

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University of Bergen, Norway, May–June 2025. Host: Prof. Fedor Fomin.

## Awards

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- Microsoft Research Travel Grant (INR 250,000), AAAI 2024.
- ACM India–IARCS Travel Grant (INR 100,000), AAAI 2024.

## Teaching and Service

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- Delivered two lectures in *Topics in Design and Analysis of Algorithms* (CS6100), IIT Madras.
- Teaching assistant for programming courses at IIT Guwahati.
- Reviewer for *Acta Informatica*, AAAI 2026, and NeurIPS 2026.

## Skills

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**Programming:** C++, Python.    **Languages:** Hindi (native), English (professional proficiency).